The size of the informal economy in Nigeria: a structural equation approach

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Abstract

Purpose – This paper contributes to the literature concerning the Nigerian informal economy (IE) by estimating its size from 1991 to 2017 and identifying the major causes.

Design/methodology/approach – A structural equation approach in the form of the multiple indicators multiple causes (MIMIC) method is used to estimate the size of the Nigerian IE.

Findings – The results indicate that vulnerable employment and urban population as a percentage of the total population are the main drivers of the IE in Nigeria. The IE in Nigeria ranges from 38.83% to 57.55% of gross domestic product (GDP).

Research limitations/implications – As a result of the empirical challenges in the estimation of the IE, the estimates of Nigeria's IE are considered to be rough estimates.

Originality/value – The authors calibrated the MIMIC model with the official estimate of the informal sector published by the Nigerian National Bureau of Statistics (NBS). This was an attempt to combine the national accounting approach, to estimate the size of IE, with the MIMIC approach, and to estimate the trend of informality.

Keywords Informal economy, Structural equation approach, MIMIC, Nigeria

Paper type Research paper

1. Introduction

It has long been argued that the presence of a large informal economy (IE) poses a hindrance to macroeconomic policies for welfare maximization and economic development. This is because a large IE impacts negatively on public finance and reduces the effectiveness of economic policies (Dell'Anno *et al.*, 2018). On the one hand, the negative impact on public finance is because unreported incomes in the IE decrease tax revenue, thereby making less fund available for refinancing and funding infrastructural development. On the other hand, the IE undermines the effectiveness of economic policies as a consequence of biased official statistics that do not capture the IE (Farazi, 2014).

Nigeria is a particularly relevant case study in this empirical literature because the estimated size of the IE is among the largest around the world. Ogbuabor and Malaolu (2013) estimated that the average size of the IE in Nigeria is about 64.6% of gross domestic product (GDP). Likewise, Schneider (2012) estimated that from 1999 to 2006, the IE in Nigeria averaged around 56.2%. The substantial role of the IE in the Nigerian economy implies that in the design of macroeconomic policy, the IE should be taken into consideration due to its linkages with the formal economy. Moreover, the IE strongly affects the exhaustiveness of the official estimates of Nigerian GDP. Awojobi *et al.* (2014) stated that the IE contributed

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significantly to the rebased size of the GDP figures from 2010 to 2014. In particular, the National Bureau of Statistics (NBS) rebased Nigeria's GDP in 2014, and the exercise revealed that the size of the official GDP was significantly underestimated. In 2013, the old official NBS estimates indicated that, in nominal terms, the GDP size was about ₦42.39 trillion (about 138.98 USD bn), while the new estimates showed that the economy was actually ₦80.22 trillion (about 263.02 USD bn; NBS, 2014). Awojobi *et al.* (2014) revealed that the IE accounted for 57.9% of Nigeria's formal GDP size.

As earlier mentioned, the IE tends to undermine the effectiveness of macroeconomic policies, especially when its size is distinctly large. Hence, for an economy like Nigeria – where the IE employs over 80% of the labor force – the design of structural policies needs to take into consideration in the IE. However, there is little empirical evidence on the trend of the Nigerian IE as existing literature usually focuses on its structural and demographic characteristics (see SMEDAN and NBS, 2013). Therefore, policymakers are limited in assessing how informality affects key macroeconomic variables, considering that they often have to rely on point estimates of the IE from surveys. In light of this, the first aim of this research is to fill this gap of the literature by estimating the size of the Nigerian IE over the period 1991–2017.

The literature identifies three main econometric approaches to estimate the IE: direct, indirect and multiple indicators multiple causes (MIMIC) approaches (Schneider and Enste, 2000). This study applies the latter by proposing a new calibration (or benchmark) approach to convert the estimates of the index of MIMIC (e.g. latent scores) to the absolute values of the informality ratio. As the benchmark approach of MIMIC method concerns, it is probably one the most controversial aspect of this application to IE. Dell'Anno and Schneider (2006) explained that the criticism is based on the difficulties faced by researchers in attempting to convert the index of the IE after estimating through the MIMIC model into cardinal values. This is a challenging task, as the latent variable, and its unit measure (e.g. unit in the local currency, million, per capita, percentage, ratios, etc.), is not observed. At this stage of research, there is not a leading view to ascertain the most appropriate benchmark method to proceed with.

Therefore, the second aim of this research consists to calibrate the IE index based on the MIMIC model with a projection of the official estimate of the sectoral composition of the IE obtained by the Nigerian NBS. This calibration approach allows taking advantage of the better performance of the statistical accounting method, to estimate the IE at a point of time and of the MIMIC model, to estimate the informality ratio over time.

The rest of the manuscript is structured as follows: Section 2 focuses on the definition and theoretical views of the IE; Section 3 documents some stylized facts on the Nigerian IE; Section 4 presents the data and methodology of the empirical analysis; Section 5 provides our estimates of the size of the Nigerian IE; Section 6 provides the discussion of findings. The conclusion is highlighted in Section 7.

2. Literature review

2.1 Definitions of informal economy, informal sector, informal employment, non-observed economy

The IE (often regarded as shadow, hidden, underground, black, unobserved economy or informal sector) has been established to be regularity around the world. Several scholars have attempted to measure its size and analyze its determinants; however, the basic question on how to define the IE is still controversial. In particular, an open debate exists between national accounts experts and applied economists on "what" the estimates of IE should include.

There have been some attempts to reconcile these two definitional approaches (e.g. Dell'Anno, 2016); however, the resolutions of the International Conference of Labor Statisticians (ICLS) published by the International Labor Organization (ILO) provide a valuable contribution to present a common definitional framework for both economists and statisticians. Specifically, the ILO (2018, p. 22) stated that the IE "is not a statistical concept but a concept for policy purposes that embodies the sum of all parts of informality." More specifically, ILO Recommendation No. 204 (ILO, 2015) clarified that the term "informal economy" referred "to all economic activities by workers and economic units that are – in law or in practice – not covered or insufficiently covered by formal arrangements."

This recommendation has become an international standard as it provides clarity as to what the IE stands for: (1) the IE does not cover illicit activities and that (2) the IE covers all workers and economic units – including enterprises, entrepreneurs and households. Explicitly, the term "economic units" in ILO's classification refers to units that (1) employ hired labor, (2) self-employed individuals and (3) cooperatives and social and solidarity economy units. Accordingly, it is relevant to point out as the concept of IE – mainly used by labor statisticians – is structurally different from the concept of non-observed economy (NOE) – introduced by the National Statistical Institute to encompass all economic activities that may not be captured in the basic data sources used for compiling national accounts. In fact, the NOE includes both legal – as underground, informal (including those undertaken by households for their own final use) – illegal and other activities omitted due to deficiencies in the basic data collection program.

Another "false friend" and source of misunderstanding among scholars in this field of research is that the labels "informal economy" and the "informal sector" have different meanings. The 15th ICLS resolution specified that the informal sector is exemplified by "units engaged in the production of goods or services with the primary objective of generating employment and incomes for the persons concerned. These units typically operate at a low level of organization, with little or no division between labor and capital as factors of production and on a small scale" (ILO, 1993, p. 7). In this sense, activities performed by economic units of the informal sector are not essentially done to evade taxes, social security contributions or infringing labor or other legislative requirements. As a result, these activities included in the informal sector should be differentiated from the activities included in the IE. A clear example of this result emerges by analyzing the relationship between informal employment – one of the most important sources of informal value-added i.e. IE – and informal sector. These two phenomena are conceptually different because while informal employment refers to the characteristics of the job [1], the concept of the informal sector refers to characteristics of production units. As a consequence, we can have formal employment in the informal sector (e.g. a formal job carried out in informal enterprise) – that should be not included in the IE – and informal employment in the formal sector (e.g. an employee of a registered enterprise whose employment relationship is, in law or in practice, not subject to national labor legislation, income taxation, social protection, etc. – that should be included in IE).

Moving from "national accounting" to "economic" definitional approach of informality, we find that the latter is usually based on the legality issues in terms of regulation and taxation. For instance, Schneider and Enste (2000, p. 79) defined the shadow economy as encompassing all "legal value-added creating activities which are not taxed or unregistered" and, in this sense, this definition is comparable to the ILO definition of IE. Mazhar and Meon (2017) explained that the IE is one that cannot be taxed because the activities are largely unrecorded and undeclared. Pratap and Quintin (2006) noted that for developing countries, the situation is more pervasive as a result of burdensome regulatory institutions.

In terms of theoretical views of the IE, Chen (2012) noted that there were four major views concerning the IE: dualist, structuralist, legalists and voluntarist. The dualist school of thought argues that the IE arises due to the imbalances between population growth and employment opportunities in the nonagricultural sector. Accordingly, if the growth rate of formal employment is lower than the growth of the urban population, we should expect an increase in the IE. The "structuralists" assert that informality arises due to excessive demand for informal labor rather than the lack of demand for formal labor as firms seek to remain profitable. Chen (2012) further noted that another factor contributing to the rise of the IE under the structuralist school of thought was the reaction of formal firms to the bargaining power of organized labor and the avoidance of state regulations, taxes and social protection of workers. The legalist and voluntarist schools of thought are similar in the sense that they both focus on workers that decide to operate informally to avoid tax and regulation burden; however, the difference between these two schools is that the voluntarists do not blame the legal framework as the reason for operating informally (Chen, 2012).

2.2 Empirical literature on the Nigerian informal economy and the multiple indicators multiple causes approach

The empirical analysis of the size of the IE in Nigeria shows that it is one of the largest economies in the world. Salisu (2001) estimated the IE from 1960 to 1997 by applying a MIMIC model with three causes (i.e. tax burden, inflation and real per capita income) and two observed indicators (i.e. male participation rate and changes in the cash demand deposit ratio). The findings indicated that the IE, as a percentage of official GDP, has been rising in Nigeria but marginally declined in the 1980s before it continued to expand in the 1990s. Schneider (2005) used both MIMIC and currency-demand approaches to estimate the size of the IE in Nigeria between 1990 and 2000. The causal variables of the MIMIC model used were the share of direct and indirect taxes as a percentage of GDP, the burden of state regulation, the unemployment and the GDP per capita. The indicator variables of the IE were the growth rate of GDP, employment as a percentage of population and changes in the currency per capita. The findings revealed that the IE in Nigeria ranged from 46.7% of GDP to 57.9% over the period 1990–2000. Schneider (2012) provided an update of his previous estimates of IE in Nigeria over the period 1997–2007. The result confirmed the previous estimates as the IE averaged about 56% of GDP.

Ogbuabor and Malaolu (2013) applied the MIMIC approach to estimate IE over the period 1970–2010. The study specified a MIMIC model in which tax burden, government consumption, unemployment, inflation, interest rate and trade openness are considered as causes of the IE, while currency in circulation (M1) and the real GDP (RGDP) index were used as indicators of IE. The study concluded that Nigeria's IE averaged about 64.6% of GDP over the period of analysis (1970–2010). Further breakdown of the results indicated that in 1970, the IE was at its highest and estimated to be 77.2% of GDP and the minimum value was calculated in 1997 (53.6% of GDP). Igudia *et al.* (2016) analyzed the determinants of the IE in Nigeria by using the MIMIC approach and the results pointed out that unemployment, poor regulatory institutions and the need to survive are among the major reasons for influencing the size of the IE in Nigeria.

3. Stylized facts on the Nigerian informal sector and informal employment

Moving from the macroeconometric approach (e.g. MIMIC) to the national accounting approach to estimate the size of the IE, often implies relying on estimates of the informal sector, which is a somewhat different phenomenon. From this viewpoint, the most detailed picture of the Nigerian informal sector is provided by the official survey of Micro, Small and Medium Enterprises (MSMEs) conducted by the Small and Medium Enterprises Development Agency of Nigeria (SMEDAN) and the Nigerian NBS in 2013.

As pointed above in Section 2, the estimates based on SMEDAN and NBS (2013) measure the size of the informal sector that does not completely overlap with the definition of IE. In particular, the informal sector includes all the microenterprises employing less than ten people and having an asset size of less than ₦5 m. According to SMEDAN and NBS (2013), microenterprises constitute 99.8% of total Nigerian MSMEs Furthermore, microenterprises had grown to 36.9 m in 2013 from 17.2 m in 2010, representing a growth rate of about 114.3% [2]. In terms of informal employment, microenterprises employ the largest proportion of the Nigerian labor force (i.e. 81.3%). These findings point out the significant role of the IE in terms of creating employment opportunities, thereby providing a source of income for Nigerians and also contributing to the economic growth of the GDP.

4. The research methodology

4.1 Estimating the informal economy by a multiple indicators multiple causes model

Structural equation modeling (SEM) is a multivariate statistical analysis technique that is used to analyze structural relationships between measured variables and latent constructs. Furthermore, while traditional techniques, such as regression analysis, allow for the use of only one single-item variable, the SEM approach allows to the inclusion of multi-item variables on both sides of the equations (Venturini and Mehmetoglu, 2019). The MIMIC model is a specific model of the SEM approach which considers the latent variable to have a relationship with a set of observable variables, regarded as indicators for changes in the size of the latent variable and also linked to some observable causal variables recognized as important influencers of the latent construct.

Referring to Dell'Anno (2007) for a broader explanation of the MIMIC approach in estimating the IE, in this study, the discussion is limited to providing the basic idea behind these estimation approaches. In short, in the SEM approach [3], the estimates of coefficients are obtained by minimizing the discrepancy between the empirical covariance matrix, *S*, and a covariance matrix implied by the model $\Sigma(\theta)$. It occurs by applying an iterative algorithm that involves repeated attempts to obtain estimates of parameters that result in the "best fit" of the model to the data, where the "best fit" means the lowest value of a discrepancy function. Several discrepancy functions are proposed, e.g. maximum likelihood (ML); generalized least squares; unweighted least squares and weighted least squares), but the (standard) approach consists to apply the Jöreskog (1967) ML discrepancy function if the hypothesis of multivariate normality in the observed data holds

$$F_{\mathrm{ML}} = \log \text{ of } \det\Sigma(\theta) + tr\left[S\Sigma^{-1}(\theta)\right] - \log \text{ of } \det(S) - (p+q) \tag{1}$$

where "det" indicates the determinant of a matrix, "*tr*" indicates the trace and $p+q$ is the total number of manifest variables (q causes and p indicators of IE) in the model. If F_{ML} is estimated on a sufficient sample size, then observed data are multivariate normality and the model is correctly specified, F_{ML} yields asymptotically correct standard errors for the parameter estimates and an overall fit statistic that follows, asymptotically, to test if the MIMIC model is correct in the population.

In the literature on the estimation of the IE by the MIMIC approach, one of the most relevant issues is on how to convert the latent scores in absolute values of IE. The common rule is to anchor the latent variable to the indicator (defined as reference indicator) measured by a suitable unit of measure (e.g. measured as real or nominal GDP, index of GDP with respect to a base year, the growth rate of GDP, etc.) such that the latent variable scores (i.e. IE) take the same scale (e.g. local currency, percentage points, per capita value, etc.). By anchoring or fixing the scale of the measurement coefficient matrix, the scholar fixes the scale of η (Stapleton, 1978). In this study, we derive a proxy of informality ratio that is used as a reference indicator of the measurement model of the latent variable. In this way, by anchoring the latent construct to a proxy of IE calculated using data on the sectoral composition of IE provided by the NBS in 2015, we attempt to integrate into our estimates, on the one hand, the greater reliability of national accounting methods to estimate the IE (i.e. this is the aim of the measurement model of the MIMIC) with the economic theory on the potential determinants of the IE for a developing country (i.e. the idea behind the structural model of the MIMIC).

4.2 Pros and cons of the multiple indicators multiple causes approach to estimate the informal economy

Dell'Anno (2007) summarized several concerns related to the accuracy of MIMIC estimates. These include the issue of the endogeneity between observed causes and indicators of the IE, the "real meaning" of the latent variable, the constraints in model specification due to convergence problems in maximum likelihood estimator and the shortcoming of using an exogenous estimate to calibrate the latent scores (i.e. benchmarking). Accordingly, MIMIC estimates should be always considered as a rough estimate of the IE. Despite this, the MIMIC approach has a relevant advantage compared to other estimation approaches: it provides a more convincing economic theory in estimating the IE than the survey method and the indirect approach (Dell'Anno *et al.*, 2018). In this sense, the trend of the IE can be estimated by exploiting economic hypotheses on the relationships between IE and macroeconomic variables. As will be explained in Sections 5.4 and 5.5, in this study, we specifically deal with the issue of benchmarking of the latent index by proposing, as reference indicator of the measurement model of the MIMIC model, a proxy of informal value added (VA) extracted by official estimates of the IE provided by the NBS (2016). Accordingly, the calibration of the MIMIC estimates is simpler, and probably more reliable, than existing alternate benchmarking methods.

5. Estimating the Nigerian informal economy

To estimate the Nigerian IE by the MIMIC approach, we apply the following five-step procedure:

5.1 Model specification: first step

Looking at the empirical literature concerning the IE (e.g. Schneider and Enste, 2000), and taking into account the peculiarities of the Nigerian IE and data availability, this study selects a set of main potential determinants and indicators of IE. Accordingly, there are six causal variables and two potential indicators of IE. The MIMIC model is defined along with the paper of Dell'Anno *et al.* (2018), where the measurement model is given as follows:

$$Y_1 = \lambda_1 \eta + \varepsilon_1 \tag{2}$$

$$Y_2 = \lambda_2 \eta + \varepsilon_2 \tag{3}$$

where Y_1 and Y_2 are the indicator variables of the IE represented by informal VA based on NBS estimates and currency ratio, respectively. The structural model is represented as follows:

$$\eta = \gamma_1 X_1 + \gamma_2 X_2 + \gamma_3 X_3 + \gamma_4 X_4 + \gamma_5 X_5 + \zeta \tag{4}$$

where X_1, X_2, X_3, X_4 and X_5 are the causal variables represented by tax burden, vulnerable employment, (female or total) unemployment rate, government expenditure as a percentage of GDP and the urban population growth rate, respectively.

5.1.1 The causal variables. X_1: The tax burden (*tax*), measured as taxless subsidies on products as a percentage of GDP, is considered as a major driver of the IE. This is because an increase in taxes causes individuals and companies to go into the IE, where tax evasion is more pronounced. From the literature, it is expected that there should be a positive relationship between the tax burden and the IE.

X_2: Vulnerable employment (*vuln_emp*) is measured as a percentage of total employment. It takes into account people who are self-employed or/and contributing family workers. These groups of individuals are defined by low-income earnings, poor working conditions and lack of job security. Therefore, the higher the vulnerable employment rate, there is a greater inclination for the IE to expand. ILO (2013) explains that "vulnerable employment can, at best, be described as a subset of informal employment," hence providing a reason why its increase can cause the IE in Nigeria to expand. Furthermore, it was identified that the vulnerable employment can be used as a proxy to represent the least proportion of workers who are vulnerable to informality.

X_3: General government final consumption expenditure as a percentage of GDP (*gov_exp*) is used to proxy the level of government involvement in the economy. We assume that a high percentage of government expenditure to GDP signifies an increasing involvement by the government in the affairs of the economy, it increases the regulatory burden and, thereby incentives individuals to go into the IE. Therefore, a positive relationship is expected between government expenditure as a percentage of GDP and the IE.

X_{4a}: The unemployment rate (*unem*) is measured as the total number of people unemployed as a percentage of the total labor force. Ogbuabor and Malaolu (2013) explained that an increase in the unemployment rate brings about a substitution effect, where workers turn to the IE in order to countervail utility losses. Ogunrinola (2011) found out that in Nigeria, because of the high levels of unemployment, many of the unemployed have had to exploit opportunities in the IE.

X_{4b}: The female unemployment rate (*unem_f*). Due to the characteristics of the informal sector, we also consider an alternate to the total unemployment rate in the form of the female unemployment rate. According to Fapohunda (2012), the informal sector employs a large number of women as a result of the constraints induced upon them in terms of family responsibilities. Hence, it can be said that an increase in the female unemployment rate will bring about an expansion of the size of Nigeria's IE.

X_5: The urban population (*urb_rate*) is measured as a percentage of the total population and reflects the proportion of the entire population living in urban areas. Therefore, it is expected that an increase in the proportion of people living in urban areas will lead to an increase in the size of the IE. This is because there will be greater demand for urban jobs over the capacity of the urban industrial sector to provide employment opportunities. Chen (2012) noted that this situation arises from the dualistic view concerning the IE as a reason for an increase in rural–urban migration.

5.1.2 The indicator variables. Y_1: Proxy of informal value added (VA) ratio. NBS (2016) estimates the VA produced by the informal sector in 2015. Given that the VA produced by the informal sector does not completely overlap with the definition of IE (see Section 3), we consider it only as an indicator of our latent construct. Accordingly, we use this proxy of the informal VA ratio as the reference indicator of the latent variable for the period 1991 to 2017.

First, we group NBS estimates for 46 economic activity sectors in seven general sectors as defined by the International Standard Industrial Classification (ISIC) of all economic activities. Accordingly, we calculate the relative dimension of informal VA by using the 2015 NBS estimates (Table 1).

In a second step, by assuming that over the period 1991–2017, the share of informal VA in each ISIC group is the same of the shares calculated by the NBS for 2015 (last column of Table 1), we estimate the informal VA by multiplying sectoral informality ratios by the official VA accounted by the NBS for the seven ISIC groups. Finally, the informality ratio is given by the ratio of total informal VA (the aggregations of the informal VA calculated for the seven ISIC groups) divided by the (official) formal VA [4].

Y_2: Currency ratio. In the IE literature, a common hypothesis is that the ratio of money outside the bank to the total money supply (M0/M2) is an important indicator reflecting the

Table 1.
Informal value added ratio by economic sectors for 2015 (NBS, 2016)

	ISIC	Sector	Informal VA (nominal local currency in millions)	Informal/formal VA
(1)	A, B	Agriculture, hunting, forestry and fishing	19,636,969	91.85%
(2)	C, E	Mining and utilities	6,742,548	0.90%
(3)	D	Manufacturing	8,973,773	12.10%
(4)	F	Construction	3,472,255	11.60%
(5)	G, H	Wholesale, retail trade, restaurants and hotels	18,922,565	55.53%
(6)	I	Transport, storage and communication	12,142,142	12.86%
(7)	J-P	Other activities (ISIC)	24,254,708	30.29%
		Total value added	*94,144,960*	*41.43%*

Source(s): Researchers' estimates based on NBS (2016)

changes in the size of the IE. Given the assumption that transactions in the IE are mainly cash-based, then changes in the ratio serve as an indicator variable for the direction of the IE, unlike the broad money that includes cash deposits in the banks. Accordingly, it is expected a positive correlation between M0/M2 and the IE due to an increase in the size of the IE will lead to an increase in the demand for cash respect to the total money supply. Figure 1 displays the path diagram of the most general MIMIC model specification (5-1-2).

5.2 The unit root analysis – separating data: second step

Since Granger and Newbold's (1974) seminal article on regressions with nonstationary variables, it is well known that estimating models in which variables are nonstationary may lead to misleading conclusions (i.e. spurious regressions). In order to take into account this issue, there is the need to transform the time series into stationary ones, by employing a difference operator or, if the variables were cointegrated, estimate an error correction model. Unfortunately, in the SEM literature, the issue of cointegration is still largely unexplored [5]; therefore, we test only for the presence of unit root and, if the time series is integrated, we deal with the possibility of obtaining spurious regressions by differencing the variables.

According to unit root tests [6], we conclude that tax burden (*tax*), vulnerable employment as percentage of total employment (*vuln_emp*), total and female unemployment rates (*unem* and *unem_f*), general government final consumption expenditure as a percentage of GDP (*gov_exp*), informal VA ratio (*inf_va*) and currency ratio (*curr_ratio*) are I(1); therefore, we first separate these time series. The percentage of urban population (*urb_rate*) is I(2); therefore, we apply second differences.

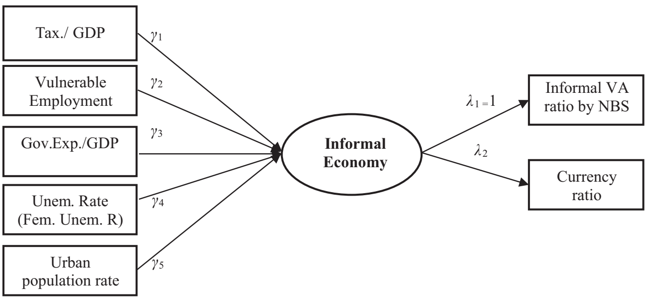

Figure 1.
The path diagram of multiple indicators multiple causes 5 (causes) – 1 (latent) – 2 (indicators)

5.3 Testing of multivariate normal distribution: third step

Following the unit root tests, we check for univariate and multivariate normality on the transformed (i.e. differenced) variables to select the appropriate discrepancy function. We find that the causal variables do not come from a univariate normal distribution, while for observed indicators (informal V.A. *ratio* and currency ratio), we cannot reject the hypothesis of univariate normality. According to Mardia, Henze-Zirkler and Doornik–Hansen tests, the data do not come from a multivariate normal distribution. Consequently, given that the ML discrepancy function compares fitted model with the saturated model under the hypothesis of multivariate normality in the observed variables, we apply the Satorra–Bentler adjustments that make standard errors, *p*-values and goodness-of-fit statistics robust to nonnormality.

5.4 Identification of the best multiple indicators multiple causes model: fourth step

The identification procedure of the best model starts from the most general specification (MIMIC 5-1-2) and continues excluding the variables which do not have statistically significant structural parameters. As often occurs in the covariance-based structural equation modeling (CB-SEM) analysis, we encounter problems of convergence of the iterative algorithm; therefore, we cannot estimate all the possible combinations of observed causes or by removing constraints on the direct effect of causes on indicators. In particular, convergence problems occur for the MIMIC specifications that include more than five observed causes (e.g. we also try to include variables that account for the institutional setting) and for the MIMIC with three causes (i.e. MIMIC 3-1-2). Table 2 reports the output of SEM estimates.

According to the goodness-of-fit statistics, the best MIMIC specification is 2-1-2. From this model, we calculate the scores for the latent variable as a linear prediction of the structural model:

$$\Delta\eta_t = 3.224\ \Delta\left(\text{vuln\_emp}_t\right) + 9.135^{*}\Delta^2\left(\text{urb\_rate}_t\right) \tag{5}$$

where Δ indicates the first differences and Δ^2 indicates the second differences.

5.5 Benchmarking or calibration: fifth step

To convert the estimated index of IE calculated by Eqn (5) into a time series of the IE as a percentage of official GDP, we propose a benchmarking that differs from the previous literature (e.g. Dell'Anno and Schneider, 2006; Dell'Anno *et al.*, 2018). This differentiation is because we use, as the *reference indicator* of the latent construct, a proxy of the IE rather than an index of official GDP as usually applied in this literature. Given that the reference indicator (informal VA) has been transformed in first differences, also the latent variable is measured in the same way, i.e. in first differences. Accordingly, we convert the estimated series of the Nigerian IE from the first differences to levels by using the exogenous estimate of informal ratio provided by NBS (2016) for the year 2015 $\text{IE}_{2015}^{\text{exog}} = 41.43\%$. Later, we apply, iteratively, the following transformations for the years before 2015, $\text{IE}_{2014}^{\text{est}} = \text{IE}_{2015}^{\text{exog}} - \Delta\eta_{2015}$; $\text{IE}_{2013}^{\text{est}} = \text{IE}_{2014}^{\text{est}} - \Delta\eta_{2014}$; etc. and for the years after 2015, $\text{IE}_{2016}^{\text{est}} = \text{IE}_{2015}^{\text{exog}} + \Delta\eta_{2016}$; $\text{IE}_{2017}^{\text{est}} = \text{IE}_{2016}^{\text{est}} + \Delta\eta_{2017}$.

Figure 2 reports the Nigerian IE estimated by MIMIC 2-1-2 over the period 1992–2017.

6. Discussion of findings

We find that the main determinants of the IE in Nigeria are the vulnerable employment and the urban population rate. The vulnerable employment captures informal workers employed in the informal sector as it is composed of both self-employed individuals and contributing family workers. The finding that vulnerable employment is a significant cause of the IE is not surprising but rather provides empirical support to the study of Ogunrinola (2011, p. 176), who

Table 2.
Multiple indicators multiple causes estimates

Causes of IE (Xq)		5-1-2a	5-1-2b	4-1-2	2-1-2
Dependent variable: informal economy					
ΔTax burden	γ_1	0.639	0.636	0.578	–
ΔVulnerable employment	γ_2	3.057***	3.064***	3.080***	3.224***
ΔGovernment expenditure/GDP	γ_3	−0.193	−0.187	−0.159	–
ΔUnemployment rate	γ_{4a}	−0.188	–	–	–
ΔFemale unemployment rate	γ_{4b}	–	−0.190	–	–
Δ^2Urban population rate	γ_5	9.188***	9.212***	9.305***	9.135***
Indicators: (*Yp*)					
Dependent variable: informal VA ratio					
ΔInformal economy	λ_1	1	1	1	1
Dependent variable: currency ratio					
ΔInformal economy	λ_2	0.082	0.081	0.785	0.062
Goodness-of-fit statistics					
Likelihood ratio					
SB χ^2 (*p*-value)		0.016	0.017	0.027	0.306□
Population error					
(pclose RMSEA < 5%)		0.303□	0.308□	0.142□	0.667□
RMSEA_adjusted_SB		0.197□	0.196□	0.209□	0.086□
Information criteria					
AIC (smaller is better)		343.28	336.56	293.98	200.97
SBC (smaller is better)		360.35	353.63	308.61	210.72
Baseline comparison					
CFI_SB		0.416	0.426	0.510	0.982□
Residuals					
SRMR		0.160	0.161	0.174	0.075
CD		0.919□	0.903□	0.869□	0.678□

Note(s): ***indicates significance at the 1% level, standard errors are robust to nonnormality; □means suitable model's fit

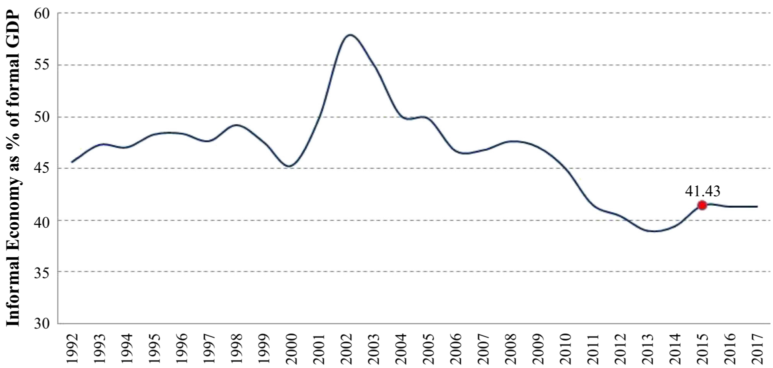

Figure 2.
Nigerian informal economy as a percentage of official (formal) gross domestic product

asserted that Nigeria's IE comprised individuals engaging in "casual and unstable employment like garbage picking, street trading, domestic help, and so on; to self-employment as master-craftsman in any given trade." As a result of this, the Nigerian IE has continued to thrive over the years, providing diverse employment opportunities to individuals seeking employment in the informal sector.

Our estimates also show that an increase in the urban population increases the size of Nigeria's IE. World Bank's (2017) data on the urban population as a percentage of the total population shows that it has increased by 42.2% in 17 years – i.e. from 34.8% in 2000 to 49.5% in 2017. This finding is in line with the theoretical views of the dualist school of thought. Specifically, the Nigerian IE has increased to absorb the excess of labor supply – due to the growing number of migrants moving to the urban areas in search of employment opportunities – compared to the labor demand of the formal economy.

Our estimates reveal that the IE has ranged from a minimum of 38.8% in 2013 to a maximum of 57.6% in 2002, while the average size of Nigeria's IE during the reference period was about 46.4% of GDP. Comparing our findings to Schneider (2012), the results indicate to some contrasting but explainable results. In particular, Schneider (2012) reported that the Nigerian IE averaged about 56.2% of GDP. An explanation for this difference could be the difference in the period. Schneider (2012) estimated the size of the Nigerian IE from 1999 to 2006 and – based on Figure 2 from our results – this coincides with the period of high and rising informality in Nigeria. However, when we compare our estimates over the same period with Schneider (2012), our findings indicate that Nigeria's IE averaged about 50.1%, which is much closer to Schneider's estimates. Another explanation for the dissimilarities between our and Schneider's estimates could emanate from the differences among structural model specifications of the MIMIC models. Schneider (2012) focused more on fiscal-related variables such as tax morale, social security indicators and quality of institutions, while our model is more focused on labor market-related determinants of the IE.

According to our findings, the most relevant increase of the Nigerian IE occurred from 2000 to 2003. The main economic cause of this increase has been the lack of inclusiveness of economic growth during this period. Looking at the data, we observe an increase in the migration of poor people from rural to urban areas to find better job opportunities. However, in the absence of a proportional increase in labor opportunities, this increase of urban labor supply has reduced working conditions and, as a consequence, increased vulnerable employment rate. In particular, from 2000 to 2003, the urban population rose from 34.8% in 2000 to 37.3% in 2003, and the number of people considered to be vulnerably employed rose from 82.4% in 2000 to 84.5% in 2003.

Our estimates show a decline in the size of IE (in terms of GDP) from 2004 to 2013. The moderation in the size of the Nigerian IE can be attributed to the various policies that the government has put in place. Okonjo-Iweala and Osafo-Kwaako (2007) opined that there was a massive civil service reform, such as the decision of the government to increase civil servant wages by 15% in January 2007. The effect was a reduction in the level of poverty. World Bank (2017) data revealed that the proportion of people living in poverty (below \$3.20 per day) marginally dropped to 77.6% in 2009 from 79.9% in 2003. More recently (2015–2017), we observe that the contribution of the Nigerian IE has remained fairly stable.

7. Conclusion

The main objective of this study is to estimate the size of the Nigerian IE from 1991 to 2017 using the MIMIC approach. The study specifies a MIMIC model with six potential causal variables (tax burden, vulnerable employment, government expenditure, female and total unemployment rate and urban population rate) and two indicators (informal VA based on NBS (2016) estimates and currency ratio) for the IE.

From a methodological perspective, this work proposes a benchmarking of the latent construct based on a reference indicator that measures a proxy of the official estimate of the Nigerian IE by the NBS for 2015 rather than an index of official GDP as the existing literature. As a result, our estimates attempt to combine the advantage of the national accounting approach to estimate the IE (i.e. reliability and exhaustiveness) with the advantage of the MIMIC approach to estimate the dynamics of the IE over time.

From a positive viewpoint, vulnerable employment and the urban population rate are the major causes of the size of the Nigerian IE. The insignificant relationship between tax burden, government expenditure as a percentage of GDP and the Nigerian IE is rather not surprising because of the peculiarities of the Nigerian economy. Tax revenue as a percentage of GDP in Nigeria is barely 2% of GDP based on data from the World Bank (2017). This implies that from the data, there is no evidence that the tax burden is a cause of the Nigerian IE. However, we have shown that the statistics on both vulnerable employment and urban population statistics in Nigeria back up the results obtained in our MIMIC results.

From a normative perspective, this result means that to reduce the size of the IE in Nigeria, the Government will have to focus on policies that address these two causes. In particular, public policies have to promote better employment opportunities to self-employments and individuals characterized by low earnings and lack of job security (i.e. the types of employed defined as "vulnerable"). As the urban population rate concerns, there is a need for infrastructural development of the rural areas in order to curb rural–urban migration. Infrastructural development in rural areas in terms of good road networks and electricity will help these areas become more attractive for investment opportunities. Investors will seek to take advantage of the infrastructural developments by expanding to the rural areas, thereby leading to the creation of employment opportunities for its dwellers and hence, the reduction of the size of the IE.

In conclusion, this study has been able to contribute to the literature on the Nigerian IE by providing a database of the contribution of the IE to GDP over the period 1991 to 2017. The estimates from this study can be used in macroeconomic models to understand how the presence of informality affects key economic aggregates (e.g. unemployment rate, business cycle, migrations, etc.).

Notes

1. According to the 17th ICLS (ILO, 2003), informal employment composes of all jobs carried out in informal enterprises as well as in formal enterprises by workers and especially employees usually without social protection or non-payment of social contribution (mainly health coverage) or the absence of written contract. At the contrary, when we refer to workers in the informal sector, we mean to comprise all people who, that were employed in at least one informal sector enterprise, irrespective of their status in employment and whether it was their main or a secondary job during a given period.

2. Small enterprises were about 68,168 and medium enterprises were just 4,670 at 2013 in Nigeria.

3. There exists two type of SEM, covariance-based (CB-SEM) and partial least squares (PLS-SEM) approaches. In this research, we consider only CB-SEM (see Bollen, 1989 for an exhaustive review of this SEM approach). Extensive reviews on the PLS approach to SEM are given in Chin (1998); Tenenhaus and Vinzi (2005); Faizan *et al.* (2018). The benefits and limitations of PLS respective to CB-SEM are still an open issue See Rönkkö *et al.* (2016) and Sarstedt *et al.* (2016) for details on this debate.

4. See Table A2 for details.

5. To the best of our knowledge, Buehn and Schneider's (2008) working paper is the only research study that deals with this issue for a MIMIC model based on SEM. Recently, Ogbuabor and Malaolu (2013) applied the Buehn and Schneider's approach to estimate IE in Nigeria. However, we opt to apply the most established MIMIC approach, due to (1) the precautionary principle to apply a method proposed by still unpublished research and (2) the sample size limitation of our data set that includes 27 annual observations that we consider reasonably inadequate to estimate a long-run equilibrium. E.g. Buehn and Schneider (2008) and Ogbuabor and Malaolu (2013) used data sets that counted more than 100 and 40 observations, respectively.

6. Details on unit root analysis are available upon request from the corresponding author.

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Appendix 1. Data sources

Symbol	Description	Source (series code)	Obs	Min	Mean	Max
Causes						
Tax	Taxless subsidies on products (current LCU) in % of GDP (current LCU)	WDI (NY.TAX.NIND.CN/ NY.GDP.MKTP.CN)	27	0.00	0.96	1.21
Unem	Unemployment, total (% of total labor force) (modeled ILO estimate)	WDI (SL.UEM.TOTL.ZS)	27	3.70	4.53	7.06
Unem_f	Unemployment, female (% of female labor force) (modeled ILO estimate)	WDI (SL.UEM.TOTL.FE.ZS)	27	3.48	4.45	6.32
Infl	Inflation, GDP deflator: linked series (annual %)	WDI(NY.GDP.DEFL.KD.ZG.AD)	27	−5.67	19.92	113.08
Vuln_emp	Vulnerable employment, total (% of total employment) (modeled ILO estimate)	WDI (SL.EMP.VULN.ZS)	27	9.48	82.17	85.38
Gov_exp	General government final consumption expenditure (% of GDP)	WDI (NE.CON.GOVT.ZS)	27	0.91	4.34	9.45
Urb_rate	Urban population (% of total)	WDI (SP.URB.TOTL.IN.ZS)	27	30.18	38.85	49.52
Indicators						
Curr_ratio	Money outside the banks as a percentage of total money supply	CBN (2017)	26	7.32	20.13	33.94
Infor_ratio	Informal total VA as percentage of formal total VA	Own calculation (see Table A2)	27	38.03	42.61	51.57

Note(s): WDI indicates World Development Indicatorsand UNSD stands for United Nations Statistics Department

Table A1. The data set to estimate multiple indicators multiple causes models

Symbols	Description	Source	Obs	Min (millions)	Mean (millions)	Max (millions)
F_agr	Formal agriculture, hunting, forestry, fishing (ISIC A-B); VA-LC	UNSD	27	123,347	7,809,550	23,952,554
F_min	Formal mining, utilities (ISIC C, E); VA-LC	UNSD	27	86,593	4,143,510	11,810,181
F_manif	Formal manufacturing (ISIC D); VA-LC	UNSD	27	28,709	2,652,857	10,044,485
F_const	Formal construction (ISIC F); VA-LC	UNSD	27	17,549	1,066,886	4,281,776
F_wholes	Formal wholesale, retail trade, restaurants, hotels (ISIC G, H); VA-LC	UNSD	27	81,446	6,355,729	22,557,932
F_trans	Formal transport, storage and communication (ISIC I); VA-LC	UNSD	27	40,151	4,158,669	13,505,047
F_other	Formal other activities (ISIC J-P); VA-LC	UNSD	27	126,643	8,074,375	28,056,199
F_totVA	Formal total; VA-LC	UNSD	27	504,439	34,261,576	113,719,049
Estimated values of informal VA (see Table 2)						
I_agr	Informal agriculture, hunting, forestry, fishing (ISIC A-B); VA-LC	Own calculation	27	113,295	7,173,072	22,000,421
I_min	Informal mining, utilities (ISIC C, E); VA-LC	Own calculation	27	780	37,325	106,386
I_manif	Informal manufacturing (ISIC D); VA-LC	Own calculation	27	3,474	320,996	1,215,383
I_const	Informal construction (ISIC F); VA-LC	Own calculation	27	2,036	123,759	496,686
I_wholes	Informal wholesale, retail trade, restaurants, hotels (ISIC G, H); VA-LC	Own calculation	27	45,227	3,529,336	12,526,419
I_trans	Informal transport, storage and communication (ISIC I); VA-LC	Own calculation	27	5,164	534,863	1,736,938
I_other	Informal other activities (ISIC J-P); VA-LC	Own calculation	27	38,355	2,445,405	8,497,100
I_totVA	Informal total; VA-LC	Own calculation	27	208,330	14,164,756	46,574,927

Note(s): UNSD stands for the United Nations, Department of Economic and Social Affairs, Statistics Division and VA-LC stands for value added in nominal local currency

Table A2. Data used to estimate the indicator of informal ratio (Y_2) based on NBS (2016)